Find the Image of a Linear Transformation

Consider $T : \mathbf{P}_3 \to \mathbf{R}^4$ where

$$T(x_0 + x_1 t + x_2 t^2 + x_3 t^3) = (y_1, y_2, y_3, y_4).$$

The description of T is given via these four mapping pairs:

$$\begin{aligned} T(1 + t + 2t^2 - t^3) &= (-1, -1, -2, -2) \\ T(-2 + t + 2t^2 + 0t^3) &= (-2, 2, -2, -1) \\ T(-3 - 0t - t^2 + 2t^3) &= (0, -1, 0, -2) \\ T(-1 - t - 2t^2 + 2t^3) &= (1, -3, 2, 1) \end{aligned}$$

Is this information sufficient for you to find the image of any $p(t) \in \mathbf{P}_4$ when the coefficients of $p(t)$ is given? We provide the information of $p(t)$ in the form of

$$p(t) = x_0 + x_1 t + x_2 t^2 + x_3 t^3$$

where the coefficient vector $x = (x_0, x_1, x_2, x_3)$ is given.

Write a function named `imageT(x)` to generate the output of this transformation.

Input Format

- $x \in \mathbf{R}^4$: The polynomial coefficient.

Output Format

- $y \in \mathbf{R}^4$: The image under the transformation T.

Implementation

Goal: Fill in the following function:

```
def imageT(x):
    ...
    return Tx
```

Make sure you import any library you use to solve the problem.